

Open-then-Audit: A Peer-to-Peer Protocol for Public Model Computation

Open-then-Audit Research

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Abstract. Repeating a large deterministic model computation at every node makes verification cost grow with the number of nodes. Delegating replay to a selected committee reduces computation but restores collusion and availability assumptions. We propose Open-then-Audit. A worker performs one exactly specified integer computation, commits its output and checkpoint trace, and makes a one-time open before receiving audit positions from a bonded threshold beacon. Nodes replay the first interval, the last interval, and a small random sample of internal intervals. In a value-bearing profile, replayers selected by future randomness from a bonded registry perform full replay and receive a fixed fee whether the result is clean or fraudulent. A false claim can be refuted by a non-interactive proof of the first divergent interval, reaching worker escrow, false-clean replayer bonds, and immature reward. Reward is proportional to actual non-padding work; padding earns zero. Security is evaluated from the cost saved by any false-interval set, beacon retry bounds, effective full-replay probability, and haircut collateral value. Sparse sampling is not a complete proof of semantic correctness, and authoritative output requires full replay or another complete proof.

1 Introduction

Rewarding public model computation requires third parties to determine whether the claimed work was performed. Full replay by every node reduces trust but repeats the same expensive computation throughout the network. Refereed delegation uses competition among servers, while Truebit, Arbitrum, optimistic rollups, and opML move expensive execution off chain through verification games or fraud proofs [4, 5, 6, 7, 8]. Open-then-Audit differs by first binding the worker to one trace, letting every node inspect a small future-random sample after opening, reducing later refutation to a non-interactive first-divergence proof, and adding protocol-funded bonded full replay. Forced errors and interactive bisection are not part of the normal path.

What is needed is a system in which the worker is bound before learning inspection positions, ordinary nodes perform only bounded computation, and a false claim later becomes a short proof without a response from the worker. Sampling formulas alone are insufficient if floating-point implementations diverge, full-attention witnesses become unbounded, replayers can remain lazy, host block hashes can be ground, or early EOS and padding can launder saved cost.

Open-then-Audit combines exact integer computation, work-proportional reward, commitment, irreversible opening, a bonded future beacon, sampled audit, bonded full replay, and a first-divergence proof. The protocol is independent of a particular blockchain, token, fork-choice rule, or model. A host provides ordering, attester finality, threshold entropy, data availability, escrow, and evidence inclusion.

2 Canonical Computation and Work Accounting

The machine description C of a value-bearing claim uniquely fixes the model or program, weights, tokenizer, input encoding, fixed-width integer or fixed-point arithmetic, requantization, rounding, operation order, routing, termination, and checkpoint rule. Floating point is forbidden in the value-bearing validity path. Public static bounds or range analysis show that every intermediate cannot overflow; an execution reaching checked overflow is invalid. The transition sequence must be independent of batch size, thread count, kernel fusion, and CUDA, ROCm, Metal, or other execution environments.

A rounding registry fixes accumulator boundaries, requantization ties, signed shifts, saturation, quantization of MoE router output, and router tie-breaking. Two independent implementations must agree bit-exactly not only on ordinary vectors but also on negative and differential corpora forcing these boundaries before a Computation Definition becomes value bearing.

Divide an execution into q intervals:

$$s_{i+1} = F_i(s_i), \quad 0 \leq i < q.$$

Every reward-bearing interval carries the same canonical work quota K . Let $q = n_w$ be the number of non-padding reward-bearing intervals; padding is excluded from the audit population, reward, and probability formulas. After canonical EOS, remaining shape contains only the zero-cost, zero-reward transition

$$D_{i+1} = H(D_i \parallel \text{PAD} \parallel i).$$

Claim reward or lottery weight is strictly proportional to actual non-padding work:

$$R_{\text{claim}} = \rho K n_w. \quad (1)$$

A worker jumping falsely to early EOS loses reward for all remaining work. To claim reward for an omitted interval, it must commit that interval as work bearing, placing it in the false-transition set S . Padding does not enter random sampling and cannot inflate q or the defect fraction. Therefore rewarded work satisfies

$$C_{\text{saved}}(S) \leq K|S|. \quad (2)$$

Accounting-only padding that counts unperformed work as completed is forbidden.

At each boundary the worker records

$$D_{i+1} = H(D_i \parallel A_i \parallel \Delta S_i \parallel U_i),$$

where A_i is an operation accumulator, ΔS_i is a changed-state commitment, and U_i includes token and layer position, KV length, append-only KV root, attention position, router state, selected experts, expert accumulation order, quantization scale, rounding, termination, padding, RNG, and every other mutable value affecting the next transition.

The worker performs one full execution and incrementally produces $D_0, \dots, D_q$. Eighteen audited intervals do not mean eighteen inferences. A second full inference, full-state rehash at every checkpoint, synchronous transfer of large tensors, and global accelerator synchronization are not security requirements.

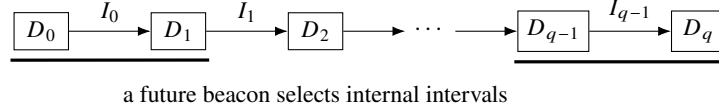


Figure 1: The worker commits to the full trace. Nodes replay boundary intervals and future-random internal intervals.

3 Commitment, Opening, and a Bonded Beacon

The worker publishes an ExecutionCommit containing an output commitment, checkpoint-vector root, vector data-availability manifest, non-padding work count, and dedicated escrow. The protocol does not advance until the commit is attester final. If the commit expires before opening, the escrow is returned except for a non-refundable attempt fee.

After commit finality, the worker publishes a one-time OpenAnnounce revealing the output, execution seed, and checkpoint-vector encoding. A second open for the same commit is invalid. Audit entropy is generated only after the OpenAnnounce becomes attester final.

A value-bearing host must not use block hashes alone as audit entropy. It provides a threshold commit-reveal beacon operated by a bonded attester set. In each round, DKG or an equivalent verifiable distributed setup threshold-shares ρ_r and finalizes one commitment $C_r = H(\rho_r)$ before Open finality. Fewer than the threshold shares cannot reconstruct ρ_r . After Open finality, unique reveal shares are published; any threshold subset of valid shares reconstructs the same unique ρ_r , so selecting a reveal subset cannot create a different output. Missing shares are objectively slashable. Signatures authenticate shares but signature bytes are not entropy.

Audit randomness is

$$R = H(\text{ClaimID} \parallel \rho_r \parallel \text{ParameterSetID}).$$

A pre-colluding threshold can predict the beacon, so its collusion bound is a host assumption. A new round may begin only if the entire round cannot reconstruct by its deadline. If adversarial attestors have total burnable bond B_{adv} and each failed round incurs at least irreversible slash s_{min} , the host publishes

$$g_{\text{max}} \leq 1 + \left\lceil \frac{B_{\text{adv}}}{s_{\text{min}}} \right\rceil$$

or a more conservative bound. A host unable to price g_{max} cannot activate a value-bearing profile.

4 Sampled Audit and KV Availability

For $q > 18$, nodes verify the first interval I_0 , the last interval I_{q-1} , and m distinct internal intervals selected from R . The reference value is $m = 16$, for eighteen intervals in total. When $q \leq 18$, all intervals are verified.

For each selected interval j , the worker supplies a state opening for D_j , operation accumulator, Merkle paths to required model tiles, and claimed D_{j+1} . A node executes F_j once and passes only if $D'_{j+1} = D_{j+1}$. The first interval binds the trace to the public job; the last binds it to output and termination state.

In full attention, a selected interval reads the entire preceding KV prefix. KV state is stored in a Merkle Mountain Range or equivalent append-only commitment. The package does not duplicate the prefix for every interval. It carries one canonical KV prefix through the maximum selected position, plus membership and consistency proofs for each sample. For context cap c_{ctx} , KV bytes per token b_{KV} , and model-tile witness W_{tile} , a profile publishes

$$W_{\text{max}} = c_{\text{ctx}} b_{\text{KV}} + W_{\text{tile}} + O(m \log c_{\text{ctx}}) \quad (3)$$

and, for data and evidence band B_{DA} and package deadline Δ_{pkg} , requires

$$W_{\text{max}} \leq B_{\text{DA}} \Delta_{\text{pkg}}. \quad (4)$$

A long-context profile failing (3)–(4) is not admitted to the sampled-audit tier and is full-replay-only. A 128k-class context remains full-replay-only until a release-specific benchmark satisfies both equations.

The complete ordered checkpoint vector, KV manifest, and sampled witnesses must be objectively data available before audit entropy and reconstructable from host history throughout the proof period. A root plus best-effort peer serving is not data availability.

5 Bonded Required Replay

Sampled audit is effective against broad omission but can miss a sparse junction. A profile with spendable reward therefore selects every Provisional claim for RequiredReplay with probability $r > 0$ from a later beacon and assigns k distinct bond outpoints from a bonded replayer registry without replacement. First-come assignment is forbidden. The worker’s identity and bond outpoint are excluded, but this does not prove Sybil independence.

Each replayer creates claim-specific bond reservation, performs canonical full replay, and signs a ReplayRoot and clean or fraud result. It receives a fixed fee in either case. Maturity requires valid completion statements for all k selected slots. If any replayer reports fraud, a DefectProof is processed and the claim cannot mature. No-show, invalid statement, and timeout cause bonded penalty and bounded reassignment. If k completions cannot be obtained within the total audit horizon, reward becomes zero and, absent worker fault, the worker escrow is returned. A later valid DefectProof also slashes replayer bonds that signed false-clean.

A ReplayStatement is not cryptographic proof of full replay and cannot cryptographically detect a lazy replayer copying the worker’s public root. Security is therefore measured by collateral exposed by the attack coalition, not signer count. For worker and colluding replayer set Q ,

$$L_{\text{coal}}(Q) = L_W + \sum_{a \in Q} L_a. \quad (5)$$

The raw assignment rate r is not counted as security. p_{full} is a conservative lower bound including selection, beacon bias, operator independence, honesty, completion, and proof inclusion: the probability that at least one non-colluding or permissionless full replay can produce a DefectProof before the deadline.

Selected replayers receive fixed fees even for clean claims. A permissionless reporter bounty covers at least measured p99 full-replay cost plus margin:

$$B_{\text{report}} \geq C_{\text{replay}, \text{p99}} + \mu. \quad (6)$$

Total reporter payment remains capped at a predeclared small fraction so self-reporting cannot recover most of the penalty; the irrecoverable coalition loss is used in the economic inequality. New value-bearing admission stops if replay capacity or bonds cannot maintain the published p_{full} lower bound.

A profile consuming authoritative model output requires full replay of every claim, equivalent to $p_{\text{full}} = 1$, or another complete proof of execution correctness.

6 Network, Fraud Proof, and Host Requirements

The network operates as follows:

1. The worker fully executes a job, commits the trace, and locks dedicated claim escrow.
2. The attestor layer finalizes the ExecutionCommit.
3. The worker publishes a one-time OpenAnnounce and makes the checkpoint vector and KV manifest available.
4. The attestor layer finalizes the open and reveals the threshold beacon.
5. The worker broadcasts sampled interval witnesses and nodes perform bounded replay.
6. A valid claim becomes Provisional.
7. A later beacon selects RequiredReplay and k replayers, and required claims wait for completion before maturity.
8. A valid DefectProof Voids the claim; otherwise only a claim satisfying all conditions becomes Mature.

A full replayer identifies the first interval j^* where the canonical sequence differs from the worker’s sequence. By minimality, the start state D_{j^*} is correct while claimed D_{j^*+1} is false. The proof contains membership paths, pre-state openings, data for one interval, and the claimed next digest. A node executes one interval. No worker response, turn clock, bisection, or operator judgment is required.

The worker may publish a timely PackageCommit binding the complete AuditPackage and an objective data-availability certificate before the deadline. No-fault return for host inclusion failure is permitted only when this commit and certificate exist and the package later verifies. A missing, unavailable, or invalid package forfeits the claim.

A reporter may publish an EvidenceCommit binding proof hash, reporter key, salt, and deposit. The first timely commit merely starts one fixed HoldEnd and receives no exclusive slot. Other reporters inside the original window may reveal through the same HoldEnd; non-reveal burns only that reporter’s deposit. A valid direct proof is accepted immediately regardless of holds.

Acceptance of a valid proof freezes worker escrow, false-clean replayer bonds, and immature reward in the same state transition. Evidence is processed before maturity when both occur in one host unit. Most value remaining after reporter payment is irreversible loss.

A Host Adapter publishes at least these consensus rules:

- attester-final Commit and Open, beacon rounds, threshold, and non-reveal slash;
- rollback before finality, failure definition for deep reorg after finality, and re-inclusion rights for timely evidence;
- an objective predicate for a full evidence band and deadline pause;
- retention rules exempting checkpoint vectors, KV prefixes, and AuditPackages from pruning until settlement finality;
- CPU, RAM, storage, disk-read, network, and proof-queue caps per object, claim, and time window.

If a correct package was data available before the deadline but only host inclusion failed, reward may become zero and worker escrow may be returned except for the attempt fee. Missing, unavailable, or invalid packages are post-open failures. OTA cannot guarantee slashing if a host majority can censor valid proofs beyond the deadline.

7 Incentive and Calculations

Let S be the false-transition set among the $q-2$ internal intervals and m the sample count. One draw detects fraud with probability

$$p_{\text{sample}}(S) = 1 - \frac{\binom{q-2-|S|}{m}}{\binom{q-2}{m}}. \quad (7)$$

If an attacker can choose the most favorable among at most $g_{\max}$ independent draws, the sampled detection lower bound is

$$p_{\text{pre}}(S) = p_{\text{sample}}(S)^{g_{\max}}. \quad (8)$$

Correlated candidates or stronger host bias use a smaller directly measured lower bound. If RequiredReplay is independent and has effective lower bound p_{full} ,

$$p_{\text{detect}}(S) = 1 - (1 - p_{\text{pre}}(S))(1 - p_{\text{full}}). \quad (9)$$

An attack requiring only the worker uses L_W ; an attack requiring colluding selected replayers uses $L_{\text{coal}}(Q)$ from (5). Let $B(Q)$ be replayer bribery cost, $L_{\min}(S, Q)$ the haircut minimum irreversible value of the attack coalition over the proof horizon, $V(S)$ maximum protocol and external value, $p_{\text{obj}}(S, Q)$ conservative objective-proof probability, and M a margin. For every allowed S and every coalition Q ,

$$p_{\text{obj}}(S, Q)L_{\min}(S, Q) > C_{\text{saved}}(S) + V(S) + B(Q) + M. \quad (10)$$

For $Q = \emptyset$, (9) may lower-bound p_{obj} . If all selected replayers are captured, only directly measured permissionless watching and evidence inclusion are counted. Nominal token quantity, reputation, and future fee income do not replace hard collateral. Haircut value must be verifiable from host state; if (10) fails, new admission stops, reward falls, or p_{full} rises.

A bounded-reward core profile caps $V(S)$ by protocol and may permit $p_{\text{full}} < 1$, but collateral efficiency degrades approximately as $1/p_{\text{full}}$ and becomes an economic throughput limit. Paid inference, oracles, bridges, finance, and other external-value profiles are placed in a $p_{\text{full}} = 1$ tier or require a complete proof.

With $m = 16$ and a 25% defect fraction, one-draw detection is at least $1 - 0.75^{16} \approx 98.998\%$. For one defective junction, $p_{\text{sample}} \approx 16/(q-2)$, but a value-bearing profile connects sparse-fraud detection to p_{full} through (9). Equation (2) binds saved cost to false-interval count.

8 Limitations and Activation

OTA inherits host finality, threshold beacon, data availability, and evidence inclusion. It does not remain secure against a threshold attester coalition, a deep reorg reversing finalized open, censorship of valid evidence beyond the deadline, or a common bug in the canonical integer VM. Replay signer count is not proof of independent computation; collusion risk is included in (5) and (10).

A value-bearing deployment is invalid until it publishes an exact-integer Computation Definition, two independent implementations and a boundary corpus, padding and reward rules, KV-MMR with numerical $W_{\max}$ and context cap, attester beacon and $g_{\max}$, bonded replayer registry, k , effective p_{full} , reporter bounty, haircut collateral, Host Adapter, and resource benchmarks. Any wrong proof, wrong slash, missing slash, or cross-implementation divergence stops new admission.

This paper alone does not establish readiness of a particular mainnet. Spendable reward is enabled only after a bounded-reward profile's Host Adapter, Computation Definition, Parameter Set, and implementation audit satisfy (3), (4), (10), and every activation gate.

9 Conclusion

Open-then-Audit binds a worker to one execution before audit positions exist and asks ordinary nodes to replay only a small number of bounded intervals. Exact integer arithmetic closes honest implementation divergence; work-proportional reward closes early-EOS and padding cost laundering. Append-only KV commitment makes full-attention witness size measurable, and a bonded threshold beacon separates audit randomness from high-BPS block-hash grinding. Bonded RequiredReplay moves sparse fraud from volunteer goodwill to a protocol parameter, while a first-divergence proof turns discovered fraud into short objective evidence.

The security claim is not that eighteen intervals prove the whole computation. Saved cost is bound to false transitions, beacon retries are priced by bond, a conservative lower bound on non-colluding full replay exists, and equation (10) holds for haircut attack-coalition loss over every attack set. Complete semantic assurance requires full replay of every claim or another complete proof. Refusing to disguise sparse sampling as universal verification preserves both simplicity and security.

References

- [1] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.
- [2] R. C. Merkle, “Protocols for Public Key Cryptosystems,” IEEE Symposium on Security and Privacy, 1980.
- [3] S. Haber and W. S. Stornetta, “How to Time-Stamp a Digital Document,” Journal of Cryptology, 1991.
- [4] R. Canetti, B. Riva, and G. N. Rothblum, “Refereed Delegation of Computation,” Information and Computation, vol. 226, pp. 16–36, 2013.
- [5] J. Teutsch and C. Reitwießner, “A Scalable Verification Solution for Blockchains,” arXiv:1908.04756, 2019.
- [6] J. Koch and C. Reitwießner, “A Predictable Incentive Mechanism for TrueBit,” arXiv:1806.11476, 2018.
- [7] H. Kalodner, S. Goldfeder, X. Chen, S. M. Weinberg, and E. W. Felten, “Arbitrum: Scalable, Private Smart Contracts,” 27th USENIX Security Symposium, 2018.
- [8] K. D. Conway, C. So, X. Yu, and K. Wong, “opML: Optimistic Machine Learning on Blockchain,” arXiv:2401.17555, 2024.